

Effect of Phosphorus on the Activity and Stability of Supported Cobalt Catalysts for Fischer-Tropsch Synthesis

Michela Martinelli,^[a] Muthu Kumaran Gnanamani,^[a] Shelley D. Hopps,^[a] Dennis E. Sparks,^[a] Aimee MacLennan,^[b] Yongfeng Hu,^[b] Burtron H. Davis,^[a] and Gary Jacobs^{*,[c]}

Phosphorus promotion on Fischer-Tropsch (FT) synthesis was investigated for Co/Al₂O₃ and Co/SiO₂ catalysts having the same Co/P ratio. When P is added to Co/Al₂O₃, CO conversion on a per g catalyst basis decreased, while methane selectivity increased. Catalyst stability was higher for the sample containing both P and Pt. The main cause for lower initial conversion is Co site blocking, while the lower extent of cobalt reduction for the P-promoted Co/Al₂O₃ sample played a lesser role. When SiO₂ is used to support cobalt particles, an initial induction period for the P-promoted catalyst was observed, where CO conversion increased. Higher CO conversion at steady state, as well as improved catalyst stability during FT testing, suggest that P hindered sintering. Over the same period, a decline and

leveling off of conversion were observed for the unpromoted catalyst. Completely different effects were observed depending on support type. P only acted as a poison for Co/Al₂O₃, whereas beneficial effects on steady state CO conversion and stability occurred with Co/SiO₂. The different effects of P for Al₂O₃ and SiO₂ supported Co catalysts can be explained by differences in Co-support interactions. With alumina, Co clusters are already stabilized by strong interactions with the support. P has no benefit, as it mainly interacts with alumina instead of Co; pore blocking by P also occurred. In contrast, SiO₂ has weak interactions with Co and less Co cluster stabilization. With P promotion, P anchors Co to the support, improving Co dispersion, stability and steady-state conversion.

Introduction

Fischer-Tropsch synthesis (FTS) on cobalt and iron-based catalysts has been widely investigated as a promising route to convert syngas (derived by coal, natural gas or biomass) to environmentally clean fuels and chemicals.^[1–3] Cobalt based catalysts are advantageous because of their relatively high FTS activity and C₅₊ selectivity, low activity for the water-gas-shift (WGS) reaction, and their resistance to deactivation.^[4–6] However, cobalt is more expensive than iron, so a good metal dispersion, optimal crystal size (typically in the range of 6–15 nm) and facile reducibility^[7–11] are needed to achieve optimal performance by improving usage efficiency. These parameters are influenced by different variables such as type of support (e.g. Al₂O₃, TiO₂ and SiO₂), promoters, and method of preparation.^[12–14]

The effect of promoters has been widely studied.^[7,12,15–17] Noble metal promoters (e.g. Ru, Pt, Re, Ag, Au) facilitate the reduction of cobalt oxide species interacting with the support to lower temperatures. The noble metal promoters catalyze cobalt oxide reduction presumably by hydrogen dissociation and spill-over from the promoter surface. However, a chemical promoting effect cannot be ruled out.^[4,12,15,16]

Textural promoters are typically added to: increase cobalt dispersion,^[5] improve attrition resistance,^[14,18] enhance sulfur tolerance,^[19] prevent strong cobalt-support interactions,^[5,20] or electronically modify the active metal site.^[12] When strong metal-support interactions occur, a high fraction of unreduced cobalt (mainly CoO), that is strongly bound to the support surface, remains and requires higher temperatures (i.e., higher than standard activation conditions) to be reduced.^[7] Structural promoters, such as group II elements (e.g. Mg,^[21–23] Ca^[24,25]), La,^[26–28] and Zn^[29] added in an optimal amount, have been proposed to inhibit the metal-support interaction and increase the cobalt reducibility, resulting in improved catalytic activity. Zirconium promotion has been widely investigated in the literature; however its effect on the cobalt reducibility and dispersion depends on several parameters.^[12,30–34] For example, Ma et al.^[34] found that the addition of Zr to narrow pore or wide pore 25%Co/Al₂O₃ catalysts affected the catalyst performance in different ways. An increase in the cobalt dispersion was obtained with the wide pore catalyst due to a narrowing of the pore (decreasing reducibility), while an increase in reducibility was found with the narrow pore catalyst, as Zr forced Co clusters closer to the pore mouth. Feller et al.^[30] showed that when zirconium was added to 8.5%Co/SiO₂, the metal-support interaction decreased, improving the reducibility. The authors

[a] Dr. M. Martinelli, Dr. M. Kumaran Gnanamani, S. D. Hopps, D. E. Sparks, Dr. B. H. Davis
Center for Applied Energy Research
University of Kentucky
2540 Research Park Dr.
Lexington KY 40511 (USA)

[b] A. MacLennan, Dr. Y. Hu
Canadian Light Source Inc.
44 Innovation Blvd.
Saskatoon, SK S7N2V3 (Canada)

[c] Dr. G. Jacobs
Chemical Engineering Program
Department of Biomedical Engineering and Department of Mechanical Engineering
University of Texas at San Antonio
UTSA Circle, AET 1.352
San Antonio TX 78249 (USA)
E-mail: gary.jacobs@utsa.edu

explained that the cobalt-zirconium species formed can be reduced at lower temperature than cobalt silicate.^[30] In a different investigation using H₂-activated air calcined 20%Co/SiO₂ catalyst, the dispersion was found to be very poor, and adding Zr increased the metal-support interaction, resulting in higher dispersions albeit with slightly lower reducibility.^[12] Among the textural promoters, phosphorus is also reported to modify the metal-support interaction. The partial formation of aluminum phosphate on γ -Al₂O₃ is proposed to improve the stability of the catalyst during FTS by suppressing the metal support interaction and by enhancing the reducibility of the metal.^[35–40] Indeed, phosphorus has a higher affinity with alumina than cobalt, and therefore, an AlPO₄ phase is formed. However, an optimal P loading is deemed necessary. Higher phosphorus concentration is suggested to lower activity because of the possible formation of inactive cobalt (e.g., cobalt phosphate)^[35] or blocking of active sites.^[41,42] Gnanamani et al.^[41] reported that the highest FTS activity was achieved with an optimal P-loading of 1 wt.% on Co/SiO₂. Further addition of P resulted in a detrimental decrease in both FT activity and heavy hydrocarbons selectivity. Furthermore, the authors reported that the addition of P up to 1 wt.% hinders the sintering rate of cobalt particles, resulting in greater stability. Finally, the sequence of addition of phosphorus and cobalt was found to influence catalytic activity.^[43] The best performance was obtained when phosphorus was added before cobalt. The authors attributed this result to a higher fraction of low-temperature reducible cobalt compared to the other P-containing cobalt catalysts.

The effect of promoters depends on the type of support because of differences in metal-support interactions. For example Jacobs et al.^[12] reported that the addition 0.5% Pt on Co/Al₂O₃ doubles CO conversion on a per gram catalyst basis, while the catalytic activity increases by a factor of 1.2 for air-calcined Co/SiO₂ catalysts. Cobalt particle size was smaller for Co/Al₂O₃ because of the strong interactions between the metal and the support. These smaller particles are difficult to reduce, and thus, platinum is added to facilitate reduction, significantly boosting the active site density of cobalt and, in turn, enhancing the activity on a per gram catalyst basis. On the other hand, unpromoted air-calcined Co/SiO₂ has higher reducibility because of weak interactions, and platinum promotion only slightly increases the activity. So, it is interesting to compare the effect of the same P/Co atomic ratio on the reducibility, particle size, catalytic activity and stability for a catalyst having strong metal-support interactions (e.g., Al₂O₃) versus relatively weaker metal support interactions (e.g., air calcined catalysts using SiO₂ as the carrier). Cobalt and phosphorus were characterized using various techniques (BET, BJH, TPR, XRD, and XANES). Finally, the combined effect of platinum and phosphorous was investigated using Al₂O₃ support to analyze for the presence of interactions between P and Pt, and determine their effect on catalyst performance.

Results and Discussion

Catalyst Characterization

ICP elemental analysis for prepared samples are reported in Table 1. The cobalt loading was 26 ± 0.3 wt.% for all the

Table 1. ICP elemental analysis for the prepared samples.

Sample ID	Sample description	Elemental analysis [wt.%]		
		Co	P	Pt
Co/Al ₂ O ₃	25 %Co/Al ₂ O ₃	26.30	–	–
Pt–Co/Al ₂ O ₃	0.5 %Pt–25 %Co/Al ₂ O ₃	26.12	–	0.44
P–Co/Al ₂ O ₃	1.25 %P–25 %Co/Al ₂ O ₃	25.92	1.06	–
P–Pt–Co/Al ₂ O ₃	1.25 %P–0.5 %Pt–25 %Co/Al ₂ O ₃	25.86	1.04	0.43

Table 2. BET surface area and porosity data from physisorption measurements of nitrogen at 77 K.

Sample ID	A _s [m ² /g]	Single Point Pore Volume [cm ³ /g]	Pore volume (BJH Des) [cm ³ /g]	Average Single point diameter [Å]	Average Diameter (BJH Des) [Å]
Co/Al ₂ O ₃	94.3	0.242	0.245	103	95
Pt–Co/Al ₂ O ₃	104.1	0.237	0.233	102	92
P–Co/Al ₂ O ₃	100.0	0.221	0.219	89	88
P–Pt–Co/Al ₂ O ₃	98.0	0.198	0.194	81	82

samples, while the phosphorus content (1 wt.%) was slightly lower than the theoretical value (1.25 wt.%) for both of the P-promoted samples. BET results are reported in Table 2. The addition of cobalt on alumina decreased BET surface area from 140.3 m²/g to 94.3 m²/g. If alumina is the primary contributor to the catalyst surface area, then assuming that 25% Co corresponds to 34% of Co₃O₄, the expected decrease would be to 92.6 m²/g. A value close to this was obtained for Co/Al₂O₃ samples, suggesting minimal pore blocking. The addition of P to the catalyst slightly increased the BET surface area to a value of 100 m²/g, while the average pore diameter decreased from 95 to 88 Å. The pore size distribution for Co/Al₂O₃ and P–Co/Al₂O₃ samples (Figure 1) revealed a main unimodal pore size centered at 100 Å. Furthermore, the distribution remained narrow, indicating that the cobalt particles do not significantly alter the porous structure of the support. Interestingly, a very small peak around 40 Å was observed for P–Co/Al₂O₃ sample. Park et al.^[35] found that mesopores at 50–60 Å are formed when the P/Al ratio is higher than 0.02. The authors attributed this to the formation of new mesopores ascribed to partially formed aluminum phosphate on the surface. In the case of Pt-promoted catalyst, the addition of phosphorous slightly decreased the surface area from 104.1 m²/g to 98 m²/g, while the average pore size diameter decreased from 92 Å to 82 Å. Although high surface area and optimal mesopore size are important parameters for achieving useful cobalt dispersions^[13,44] (e.g., 6–15 nm), the small morphological variation observed in the prepared samples should not account for large changes in catalytic activity and stability during FTS.

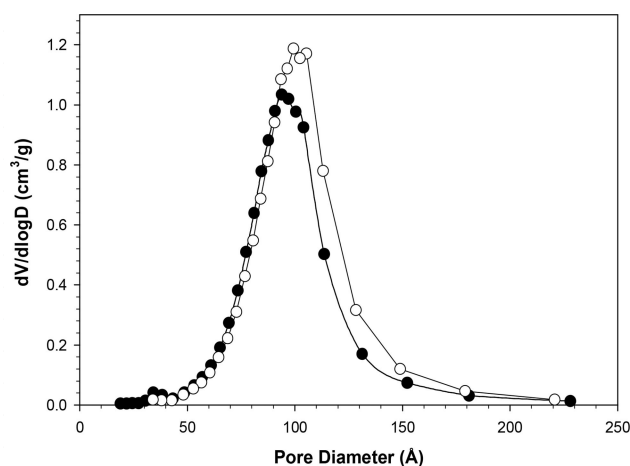


Figure 1. BBJ pore size distribution (desorption branch) for Co/Al₂O₃ (empty circle) and P-Co/Al₂O₃ (full circle) samples.

Hydrogen chemisorption with pulse reoxidation allowed for estimates of the cobalt dispersion and thus, the average cobalt crystallite cluster size. The results are reported in Table 3. The percentage reduction of cobalt decreased from 39.5% to 34.2% with the addition of phosphorus, while it only decreased from 65.6 to 64.1% when platinum was added to the sample. Pt promoter enhances the reducibility of cobalt as it allows the reduction of smaller Co surface species that interact with the support.^[12] Therefore, the negative effect of phosphorous on reducibility was mitigated in part by platinum addition. Furthermore, phosphorus promotion did not influence the particle size as the diameter was around 10 nm for both Pt-Co/Al₂O₃ and P-Pt-Co/Al₂O₃ samples, whereas the diameter slightly decreased from 19.2 nm (Co/Al₂O₃) to 17.4 nm (P-Co/Al₂O₃). The addition of phosphorous hindered hydrogen desorption, as TPD profiles were shifted to higher temperature (data not shown).

The XRD patterns are reported in Figure 2. All the catalysts show the characteristic reflection peak at $2\Theta = 36.8^\circ$ due to the presence of Co₃O₄. The particle sizes of Co₃O₄, calculated by using the Scherrer equation ($2\Theta = 36.8^\circ$), are reported in Table 4. The size of the metallic cobalt cluster was calculated by considering a contraction factor of 0.748. The values obtained by XRD were lower than those calculated by H₂ chemisorption, but the same trend is reported. This is because XRD measures crystallite size, while chemisorption measures crystallite cluster size (e.g., reported as grapelike agglomerates).^[45]

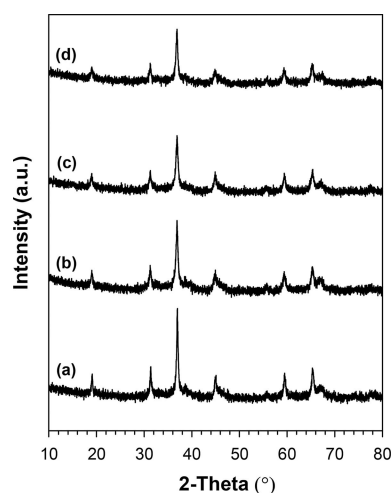


Figure 2. XRD patterns for (a) Co/Al₂O₃ (b) P-Co/Al₂O₃ (c) Pt-Co/Al₂O₃ (d) P-Pt-Co/Al₂O₃.

Table 4. Estimation of cobalt crystallite size using XRD and H₂-TPD and pulse re-oxidation technique.

Sample ID	Co ₃ O ₄ cluster size from XRD [nm]	Calculated Co ⁰ cluster size from XRD [nm]	Co ⁰ cluster size from H ₂ -TPD/reoxid. [nm]
Co/Al ₂ O ₃	19.9	14.8	19.2
Pt-Co/Al ₂ O ₃	12.3	9.1	10.2
P-Co/Al ₂ O ₃	13.3	10.0	17.4
P-Pt-Co/Al ₂ O ₃	12.5	9.3	10.1

H₂-TPR was carried out to investigate the effect of the phosphorous on the reducibility (Figure 3). TPR profiles of the cobalt alumina catalysts reveal two characteristic peaks with maxima centered around 330 °C and 650 °C. The first peak is associated with the reduction of Co₃O₄ to CoO, while the second is assigned to the reduction of CoO to metallic cobalt, as confirmed by TPR-EXAFS/XANES experiments.^[15] The addition of P shifted the maximum of the first reduction peak from 310 °C to 350 °C. This suggests that more strongly interacting Co₃O₄ particles exist when P is added to Co/Al₂O₃. In contrast, the second peak was shifted to lower temperature in the presence of P (maximum was at 578 °C for P-Co/Al₂O₃ sample, while it was at 554 °C for Co/Al₂O₃ sample), suggesting more strongly interacting CoO with addition of P. The shoulder around 700 °C for both systems was assigned to smaller, more strongly interacting CoO species. An even higher temperature

Table 3. Results for H₂-chemisorption and pulse re-oxidation.

Sample ID	H ₂ desorbed per g _{cat} [μmol/g _{cat}]	Uncorr. Disp. [%]	Uncorr. Diam. [nm]	O ₂ uptake per g _{cat} [μmol/g _{cat}]	Eff. Red. Co ₃ O ₄ [%]	Eff. Corr. Disp. [%]	Eff. Corr. Diam. [nm]
Co/Al ₂ O ₃	45.0	2.1	48.6	1119	39.5	5.4	19.2
Pt-Co/Al ₂ O ₃	140.4	6.6	15.6	1858	65.6	10.1	10.2
P-Co/Al ₂ O ₃	43.0	2.0	51.0	968	34.2	5.9	17.4
P-Pt-Co/Al ₂ O ₃	138.5	6.5	15.8	1813	64.1	10.2	10.1
Co/SiO ₂	29.5	1.7	59.3	1505.6	66.6	2.6	39.5
P-Co/SiO ₂	40.3	2.4	43.4	1148.0	50.7	4.7	27.0

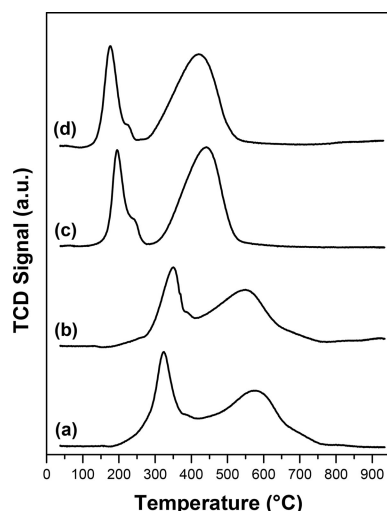


Figure 3. Temperature programmed reduction (a) Co/Al₂O₃ (b) P-Co/Al₂O₃ (c) Pt-Co/Al₂O₃ (d) P-Pt-Co/Al₂O₃.

peak, was assigned to reduction of cobalt aluminate or difficult to reduce cobalt oxide / phosphate species.

The addition of platinum shifted the peaks to lower temperature, presumably due to spillover of hydrogen from the metallic promoter to both cobalt oxide and the cobalt surface species. In this case, the presence of phosphorous shifted both peaks to lower temperature.

P-K edge spectra of various freshly reduced cobalt alumina catalysts are reported in Figure 4. The standard samples such as cobalt phosphate (Co₃(PO₄)₂) and P-doped alumina (1.25%P/Al₂O₃) are also included for the purpose of comparison. All the samples show one sharp white line peak located at 2.152 keV. This peak is due to the electronic transition from 1 s core level

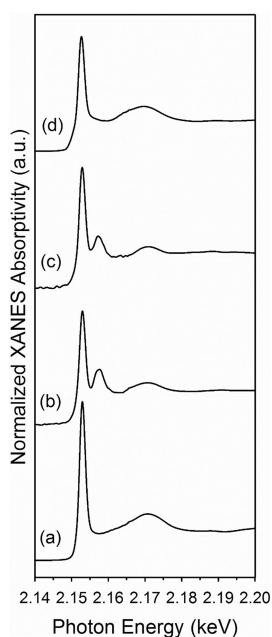


Figure 4. P K-edge XANES spectra for (a) 1.25%P/Al₂O₃ (b) P-Co/Al₂O₃ (c) Pt-Co/Al₂O₃ (d) Co₃P₂O₈.

to unoccupied 3p states. The post-edge features are assigned to multiple scattering.^[46] Interestingly, the secondary peaks centered at approximately 2.160 - 2.170 keV observed for P-Co/Al₂O₃ and P-Pt-Co/Al₂O₃ samples are attributed to the interactions between phosphorous and cobalt, as well as phosphorous and alumina. Al₂PO₄ P-K edge spectra show a characteristic post edge feature at 2.165 keV.^[47] A pre-edge peak was only clearly observed for the Co₃P₂O₈ sample. The pre-edge is due to the hybridization of P 3p states with the 3d states of the cobalt. Therefore, the lack of pre-edge for 1.25%P/Al₂O₃ is consistent with the absence of d-orbitals. Interestingly, P-Co/Al₂O₃ and P-Pt-Co/Al₂O₃ catalysts do not have a well-defined pre-edge feature, as observed previously with P-Co/SiO₂.^[43] This result suggests that phosphorus primarily interacts with alumina, and significantly less so with cobalt.

A detailed characterization of Co/SiO₂ catalysts has been previously reported.^[41,43] Briefly, the addition of phosphorus slightly decreased the surface area, while not altering significantly the pore size distribution. The cobalt reducibility and dispersion were remarkably influenced by the phosphorus promotion. Interestingly, small cobalt crystallites (Table 3) having lower reducibility were reported for P promoted Co/SiO₂ catalyst. During H₂-TPR, the low-temperature reduction peaks shifted to higher-temperature and their intensity decreased when phosphorus was added. Furthermore, two additional peaks were observed between 700 and 850 °C, which can be attributed to the reduction of cobalt phosphate or cobalt silicate. These results suggest that phosphorus enhanced the interaction between cobalt and the modified support. This is also confirmed by the pre-edge feature in P K-edge XANES as phosphorus directly interacts with cobalt through P-O-Co bonds. On the contrary, no well-defined pre-edge feature was observed for P-promoted Co/Al₂O₃.

Catalytic Activity

The evolution of CO conversion with time on-stream (T.o.S.) is shown in Figure 5. In the case of Co/Al₂O₃ catalysts, the addition of phosphorus dramatically decreased CO conversion, which dropped from 43% to 23%. This effect was more moderate when platinum was added, as the CO conversion decreased by a factor of 1.4 instead of a value of 1.9. Phosphorus deactivation was previously reported by Bradford et al.^[42] The authors suggested that phosphorus probably blocks a fraction of cobalt surface. Our results may provide support for this hypothesis, as the initial CO conversion is much lower when cobalt catalysts are promoted with phosphorus, suggesting that some cobalt active sites are lost. In fact, the small difference observed in the cobalt particle size cannot explain the difference in the catalytic activity. Park et al.^[35] reported that the phosphorus improves CO conversion at steady-state conditions when the loading is in the range of 0.5–1.0 mol%. However, their catalyst preparation is different and this can impact catalyst activity. Indeed, the authors prepared calcined Al₂PO₄ support which was impregnated with cobalt. In our case, the phosphorus and cobalt were impregnated on Al₂O₃, and the material obtained was finally

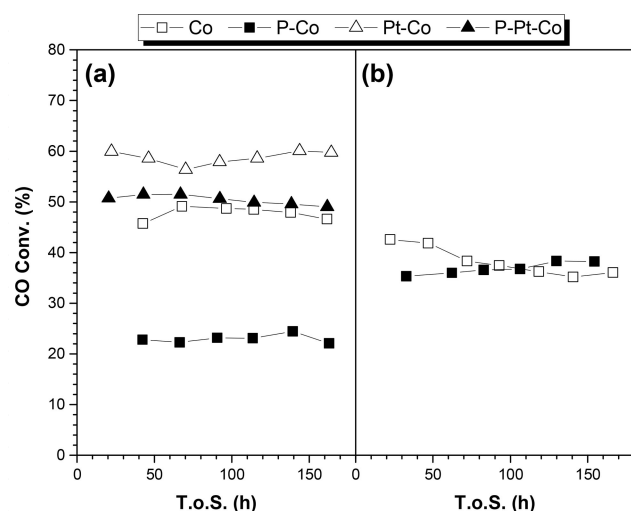


Figure 5. Evolution with T.o.S. of CO conversion for the undoped and P-doped samples supported on (a) Al_2O_3 and (b) SiO_2 ($T = 220^\circ\text{C}$, $P = 300$ psi, $\text{H}_2/\text{CO} = 2$ v/v, $\text{SV} = 15$ slph/ g_{Co}).

calcined. So, it is reasonable that phosphorus can cover some cobalt active sites in our case, especially because Al_2O_3 has a very narrow pore size distribution.

The phosphorus deactivation is less marked for the Pt-promoted sample. Probably, site blocking is not the only reason for the deactivation. The difference in cobalt reducibility impacts the activity on a per gram catalyst basis. The FTS reaction rate is a function of the number of surface metal cobalt sites, which depend on both the extent of cobalt reduction and average cobalt particle size.^[48] H_2 -TPR and the H_2 chemisorption data show that cobalt is less reducible for P-Co/ Al_2O_3 than for Co/ Al_2O_3 .

Reaction rate and turnover frequency (TOF), which is defined as molecules of CO converted per surface cobalt atom per second, are reported in Table 5. TOF values are typically

Sample ID	T.o.S. [h]	r [mol/h/ g_{cat}]	TOF [s^{-1}]
Co/ Al_2O_3	163	2.6E-2	2.9E-02
Pt-Co/ Al_2O_3	164	3.3E-2	2.1E-02
P-Co/ Al_2O_3	162	1.2E-2	1.3E-02
P-Pt-Co/ Al_2O_3	162	2.7E-2	1.7E-02
Co/ SiO_2	166	1.6E-2	5.0E-02
P-Co/ SiO_2	154	1.7E-2	2.9E-02

found to be in the range of $1.5\text{--}3.0\text{E-}02$ s^{-1} for Co supported on Al_2O_3 , SiO_2 and TiO_2 .^[13] The calculated TOF values for the catalysts used in this work were in line with the values reported in the open literature. TOF values of $2.9\text{E-}02$ s^{-1} and $2.1\text{E-}02$ s^{-1} are reported for Co/ Al_2O_3 and Pt-Co/ Al_2O_3 , respectively. TOF values were not significantly altered by the addition of phosphorus because of the structurally insensitive nature of cobalt crystallites larger than 6 nm.^[8–11,13] A similar TOF trend for P promotion was also reported by Park et al.^[38]

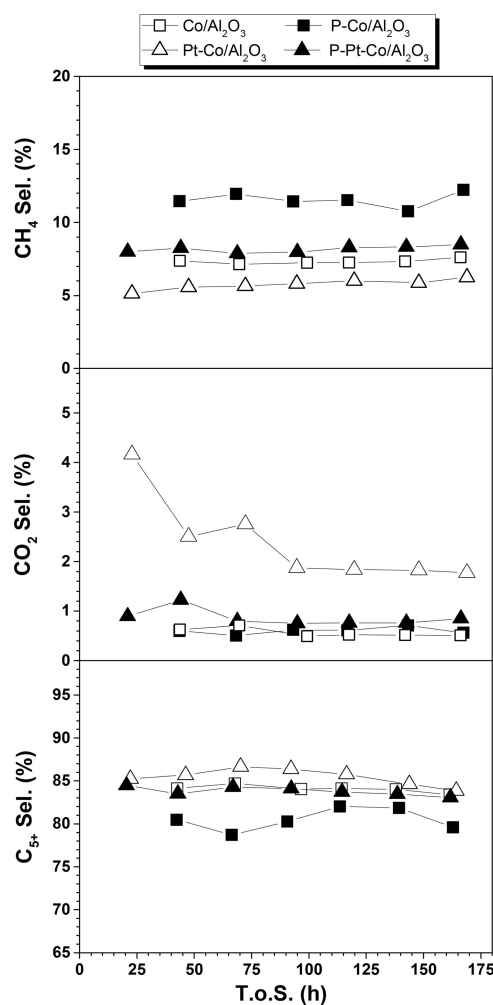


Figure 6. Evolution with T.o.S. of C-selectivity for the alumina undoped and P-doped samples (process conditions: $T = 220^\circ\text{C}$, $P = 300$ psi, $\text{H}_2/\text{CO} = 2$ v/v, $\text{SV} = 15$ slph/ g_{Co}).

The product selectivity was stable with the T.o.S. for all the catalysts (Figure 6). CO_2 selectivity for Pt-Co/ Al_2O_3 was the only exception, as it slightly decreased from 4% to a steady-value of 2% in the first 75 h. The methane selectivity was higher for the P-promoted samples. However, the CH_4 selectivity depends on conversion. So, we have compared the performance at the same conversion level (Table 6) where CH_4 selectivity was higher for P-promoted samples. TPD results show that phosphorus inhibits hydrogen desorption. Thus, a higher H/C ratio may occur during the reaction, favoring chain termination. This is in line with the higher methane selectivity, lower C_{5+} .

Sample ID	CO conversion [%]	Usage Ratio [–]	C-selectivity [%]		
			CH_4	CO_2	C_{5+}
Co/ Al_2O_3	43.0	2.2	8.1	0.5	81.6
Pt-Co/ Al_2O_3	42.2	2.1	7.6	0.7	84.3
P-Co/ Al_2O_3	40.6	2.3	10.5	0.9	81.2
P-Pt-Co/ Al_2O_3	42.1	2.2	9.8	0.9	80.6

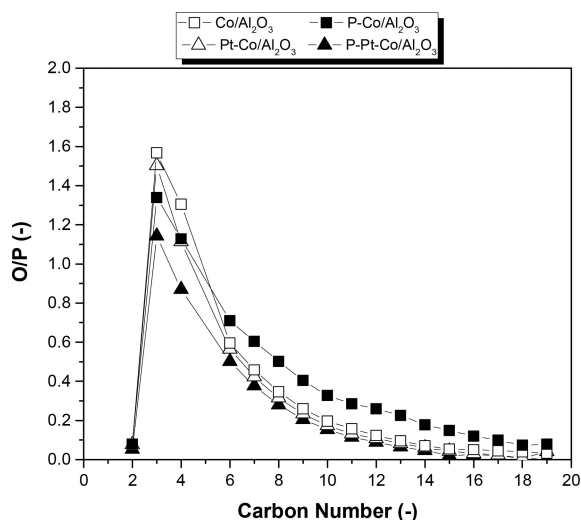


Figure 7. Olefin to paraffin ratio for alumina undoped and P-doped samples (process conditions: $T = 220^{\circ}\text{C}$, $P = 300$ psi, $\text{H}_2/\text{CO} = 2$ v/v, $\text{SV} = 15$ slph/g_{Co}).

selectivity, and lower O/P ratio at low carbon numbers (Figure 7). Surprisingly, the O/P ratio is higher for P-Co/Al₂O₃ than for the unpromoted catalyst when the carbon number is higher than 6.

The effect of phosphorus was also investigated on Co/SiO₂ systems. The same P/Co ratio was used in order to compare the activity data. The initial CO conversion for P-Co/SiO₂ was 33% and then it increased until a steady-state value of 38% was reached, while CO conversion decreased from 40% to 35% for the Co/SiO₂ catalyst. Reaction rates and TOF values at steady state conditions are reported in Table 5. The addition of phosphorus slightly decreases the TOF, as previously observed with the alumina supported Co catalysts. The phosphorus improved the stability and hindered the sintering rate of cobalt particles as the cobalt particles interact with PO₄³⁻. Gnanamani et al.^[41] reported that optimal loading was 1 wt.% for 12%Co/SiO₂. Further increases in the loading resulted in Co site blocking. The catalytic activity decreases by a factor 2.6 when the phosphorus loading is increased from 1 to 3%. We have observed that Co site blocking occurs at lower loading on Al₂O₃ than SiO₂. The two supports have different surface areas, porosity characteristics, as well as metal-support interactions. These parameters tempered the ability of phosphorus to aid in anchoring the cobalt clusters. In the case of a weakly interacting support, such as SiO₂, cobalt particles interacted with P as demonstrated by XANES and DRIFTS^[41] results. Moreover, the decrease in cobalt cluster diameter by a factor 1.4 (Table 3), a higher stability, and a higher steady-state CO conversion in FTS were observed for the P-promoted sample. These results indicate that phosphorus stabilizes cobalt clusters by anchoring them to the support. In the case of Co/Al₂O₃, cobalt clusters were already stabilized by the strong interaction with alumina. In fact, no leveling off was observed for the unpromoted Co/Al₂O₃ sample, suggesting that phosphorus played no additional benefit in anchoring the cobalt, as Co-Al₂O₃ interactions were stronger than Co-PO₄³⁻ interactions.

Conclusions

The addition of phosphorus on Co/Al₂O₃ slightly decreased Co particle size and the reducibility of cobalt oxide species during activation. This suggests that strong interactions exist between cobalt and the support. Similar results were also obtained with Co/SiO₂. Furthermore, results of P K-edge XANES analysis reveal that P favors an interaction with alumina for Co/Al₂O₃, while P interacts more so with cobalt for Co/SiO₂. The effect on the catalytic activity is completely different. An improvement in the stability was observed such that while CO conversion tended to slightly increase for the P-doped sample (suggesting an induction period), CO conversion exhibited a typical initial decline and leveling off for the undoped sample. Thus, higher CO conversion was realized once the leveling off period was achieved for the P-doped Co/SiO₂ sample, whereas a poisoning effect was observed for cobalt supported on alumina at the same Co/P ratio. The latter is due primarily to blocking of the active cobalt surface by phosphorus. The responses between the two systems could be due to differences in cobalt-support interaction. In the case of Co/Al₂O₃ sample, the cobalt clusters were already stabilized by the strong interaction with alumina. So, the addition of phosphorus has no beneficial effect. Moreover, phosphorus mainly interacts with Al₂O₃ as shown by XANES. In the case of weak metal-support interactions, such as observed with air-calcined Co/SiO₂ catalysts, cobalt clusters were not stable, as initial decline and levelling off periods were observed prior to achieving steady-state conditions. Consequently, the addition of P promoter anchored the cobalt to the support, in turn enhancing Co dispersion, stability and conversion during FT synthesis.

Experimental

Catalyst Preparation

γ -Al₂O₃ (Sasol-Catalox, 140.3 m²/g) and SiO₂ (PQ-CS2133, 327 m²/g) were used as supports for the preparation of FTS catalysts. γ -Al₂O₃ was calcined in static air at 350 °C for 4 h, while SiO₂ was calcined at 350 °C for 6 h. In the case of cobalt alumina catalysts, two phosphorous promoted samples 1.25%P-25%Co/Al₂O₃ (identified as P-Co/Al₂O₃) and 0.5%Pt-1.25%P-25%Co/Al₂O₃ (identified as Pt-P-Co/Al₂O₃) were prepared. Briefly, orthophosphoric acid (H₃PO₄, 85%, Alfa Aesar) was added by incipient wetness impregnation (IWI) in order to achieve 1.25 wt.%, and then the samples were dried overnight at 120 °C. Afterward, 25% of cobalt was added by the slurry impregnation method using a solution of cobalt nitrate hexahydrate (Co(NO₃)₂·6H₂O, Alfa Aesar). The ratio of the volume of loading solution to the weight of alumina was 1:1 as reported in a Sasol patent.^[49] Two impregnation steps were required to achieve the desired amount, and the catalysts were dried under vacuum in a rotary evaporator at 80 °C between each step (slowly increasing temperature to 100 °C). A platinum promoted catalyst was also prepared. In this case, 0.5 wt.% Pt was added by IWI with a solution of tetraamineplatinum(II) nitrate (Alfa Aesar), and the catalyst was dried overnight at 120 °C. Finally, the catalysts were calcined at 350 °C for 4 h under air flow. For the purpose of comparison, two cobalt reference catalysts, 25%Co/Al₂O₃ (named Co/Al₂O₃) and 0.5%Pt-25%Co/Al₂O₃ (named Pt-Co/Al₂O₃), were prepared by following the same procedure excluding phosphorous addition.

1%P-20%Co/SiO₂ (designated P-Co/SiO₂) and 20%Co/SiO₂ (designated Co/SiO₂) were prepared by following the same procedure, and additional details are found elsewhere.^[43] In order to shed light on the effect of the support, two different supports (Al₂O₃ and SiO₂) were utilized, and the amount of phosphorus was added to provide the same P/Co atomic ratio.

Catalyst Characterization

The nominal composition of the samples was determined by inductively coupled plasma optical emission spectroscopy (ICP-OES) using a Varian 720-ES analyzer. The catalysts were dissolved in a perchloric/nitric acid mixture. Afterwards, the spectra of the dissolved compounds (platinum, cobalt and phosphorous) were compared to those of known standard solutions.

The textural properties of the samples were determined by a Micromeritics TRISTAR 3000 analyzer. The adsorption-desorption isotherms were obtained with nitrogen at −196 °C. In each experiment, the samples (approximately 250 mg) were evacuated for 12 h at 160 °C before conducting the nitrogen adsorption measurements. The Brunauer-Emmett-Teller (BET) method was used to determine the specific surface areas of the samples, while the Barrett-Joyner-Halenda (BJH) method was used to estimate the pore volume, average pore diameter, and pore size distribution (utilizing the desorption branch).

Powder X-ray diffraction measurements were carried out using a Philips X'Pert diffractometer with monochromatic Cu K α radiation (λ = 1.54 Å). The conditions employed included a 2 θ range of 10–80°, a scan rate of per 0.01° step, and a scan time of 4 s per step.

Catalyst reducibility was measured by temperature programmed reduction in H₂ (H₂-TPR) using a Zeton-Altamira (AMI-200) instrument. The catalysts were first pre-treated in flowing argon at 350 °C for 1 h to remove the traces of adsorbed water and then cooled to room temperature. Then, the flow was switched to a mixture of 10 vol.% H₂ in argon (30 cm³/min) and the reactor temperature was increased from 25 °C to 900 °C at 10 °C/min. Hydrogen consumption was monitored by a thermal conductivity detector (TCD).

Hydrogen chemisorption was carried out using a Zeton-Altamira (AMI-200) instrument. Catalysts were first reduced in a mixture of 33% H₂ in argon (30 cm³/min) at 350 °C for 10 h and then the temperature was cooled to 100 °C under flowing hydrogen. Then, the flow was switched to argon to remove and/or prevent the adsorption of weakly bonded species. Finally, hydrogen TPD was carried out by increasing the temperature until 350 °C, and evolved H₂ was monitored by a TCD. After H₂-TPD, the sample was reoxidized at 350 °C by pulses of pure oxygen in helium,^[50] and consumption of O₂ was followed by the TCD until complete breakthrough occurred. The extent of reduction was calculated by assuming that the metallic cobalt was reoxidized to Co₃O₄. Additional details on the procedure can be found elsewhere.^[12]

The electronic structure of P was investigated by X-ray Absorption Near Edge Spectroscopy (XANES) near the P K-edge (fluorescence mode) at the Soft X-ray Microcharacterization Beamline (SXRMB) at the Canadian Light Source, Inc. To prepare the samples for XANES experiments, the catalysts were pressed into a flat pellet inside a 1" ID reactor with boron nitride. The catalyst was reduced in hydrogen at 350 °C for 10 h (heating rate 1.6 °C/min) and then fixed in wax. Finally, the reactor was cooled to 220 °C under hydrogen flow. The spectra (in energy space) were background subtracted and normalized with a two-polynomial method with degree 1 both for pre- and post-edge regions (P K-edge).

Catalyst Testing

Activity tests were carried out in a lab scale rig equipped with a 1 L continuously stirred tank reactor (CSTR). In a typical run, 10 g of calcined catalyst (63–106 μ m) were loaded into a fixed bed reactor for ex-situ reduction at 350 °C for 20 h, feeding H₂/He (1:3 v/v) mixture (60 NL/h) at atmospheric pressure. The reduced catalyst was transferred to a 1 L CSTR containing 310 g of melted Polywax 3000 by pneumatic transfer under the protection of nitrogen. The transferred catalyst was reduced in situ at 230 °C and at atmospheric pressure overnight feeding pure H₂ (30 NL/h). In this work, the process conditions utilized were: T = 220 °C, P = 20.6 bar, H₂/CO = 2 mol/mol and a stirring speed of 750 rpm. The unconverted reactants and the products leaving the reactor were sent to two traps, a warm trap maintained at 100 °C and a cold trap held at 0 °C. The uncondensed vapor stream was reduced to atmospheric pressure. The outlet gas flow was measured using a wet test meter and analyzed by an online micro-GC equipped with a TCD. The reaction products were collected in three traps maintained at different temperatures: a hot trap (200 °C), a warm trap (100 °C), and a cold trap (0 °C). The products were separated into different fractions (rewax, wax, oil, and aqueous) for quantification. The oil and light wax fractions were mixed before being analyzed with an Agilent 7890 GC equipped with DB-5 column (60 m $\times$ 0.32 mm $\times$ 0.25 μ m, Agilent J&W) and FID, while heavy waxes were analyzed with an HP 6890 GC equipped with ZB-1HT column (30 m $\times$ 0.25 mm $\times$ 0.10 μ m, Zebron) and FID.

Conflict of Interest

The authors declare no conflict of interest.

Keywords: Fischer-Tropsch synthesis • cobalt catalyst • phosphorous • alumina • silica

- [1] M. E. Dry, *Appl. Catal. A* **1996**, *138*, 319–344.
- [2] G. P. van der Laan, A. A. C. M. Beenackers, *Catal. Rev.* **1999**, *41*, 255–318.
- [3] M. E. Dry, *J. Chem. Technol. Biotechnol.* **2002**, *77*, 43–50.
- [4] A. K. Khodakov, W. Chu, P. Fongarland, *Chem. Rev.* **2007**, *107*, 1692–1744.
- [5] F. Morales, B. M. Weckhuysen (2006) In: *Catalysis* (J. J. Spivey, K. M. Dooley, eds) vol 19, RSC, Cambridge, pp 1–40.
- [6] H. Jahangiri, J. Bennet, P. Mahjoubi, K. Wilson, S. Gu, *Catal. Sci. Technol.* **2014**, *4*, 2210–2229.
- [7] G. Jacobs, W. Ma, B. Davis, *Catalysts* **2014**, *4*, 49.
- [8] G. L. Bezemer, J. H. Bitter, H. P. C. E. Kuipers, H. Oosterbeek, J. E. Holewijn, X. Xu, F. Kapteijn, A. J. van Dillen, K. P. de Jong, *J. Am. Chem. Soc.* **2006**, *128*, 3956–3964.
- [9] J. P. den Breejen, P. B. Radstake, G. L. Bezemer, J. H. Bitter, V. Froseth, A. Holmen, K. P. de Jong, *J. Am. Chem. Soc.* **2009**, *131*, 7197–7203.
- [10] P. Van Helden, I. M. Ciobica, R. L. J. Coetzee, *Catal. Today* **2016**, *261*, 48–59.
- [11] P. Munnik, P. E. de Jongh, K. P. de Jong, *J. Am. Chem. Soc.* **2014**, *136*, 7333–7340.
- [12] G. Jacobs, T. K. Das, Y. Zhang, J. Li, G. Racoillet, B. H. Davis, *Appl. Catal. A* **2002**, *233*, 263–281.
- [13] E. Iglesia, *Appl. Catal. A* **1997**, *161*, 59–78.
- [14] A. Y. Khodakov, *Catal. Today* **2009**, *144*, 251–257.
- [15] G. Jacobs, Y. Ji, B. H. Davis, D. Cronauer, A. J. Kropf, C. L. Marshall, *Appl. Catal. A* **2007**, *333*, 177–191.
- [16] F. Diehl, A. Y. Khodakov, *Oil Gas Sci. Technol. – Rev. IFP* **2009**, *64*, 11–24.
- [17] T. Jermwongratanaichai, G. Jacobs, W. D. Shafer, V. R. R. Pendyala, W. Ma, M. K. Gnanamani, S. Hopps, G. A. Thomas, B. Kitiyanan, S. Khalid, B. H. Davis, *Catal. Today* **2014**, *228*, 15–21.
- [18] D. Wei, J. G. Goodwin Jr., R. Oukaci, A. H. Singleton, *Appl. Catal. A* **2001**, *210*, 137–150.

- [19] R. V. Belosludov, S. Skahara, K. Yajima, S. Takami, M. Kubo, A. Miyamoto, *Appl. Surf. Sci.* **2002**, *189*, 245–252.
- [20] B. Cornils, W. A. Herrmann, R. Schlögl and C. H. Wong, *Catalysis from A to Z, A concise encyclopedia*, Wiley-VCH, Weinheim, **2000**.
- [21] Y. Zhang, H. Xiong, K. Liew, J. Li, *J. Mol. Catal. A: Chem.* **2005**, *237*, 172–181.
- [22] A. Guerrero-Ruiz, A. Sepulveda-Escribano, I. Rodriguez-Ramos, *Appl. Catal. A* **1994**, *120*, 71–83.
- [23] K. Shimura, T. Miyazawa, T. Hanaoka, S. Hirata, *Appl. Catal. A* **2015**, *494*, 1–11.
- [24] A. Bao, L. Liew, J. Li, *J. Mol. Catal. A: Chem.* **2009**, *304*, 45–581.
- [25] A. R. de la Osa, A. De Lucas, A. Romero, J. L. Valverde, P. Sanches, *Fuel* **2011**, *90*, 1935–1945.
- [26] Z. Cai, J. Li, K. Liew, J. Hu, *J. Mol. Catal. A* **2010**, *330*, 10–17.
- [27] C. Brabant, A. Khodakov, A. Griboval-Constant, *C. R. Chim.* **2017**, *20*, 40–46.
- [28] T. Wang, Y. Ding, Y. Lu, H. Zhu, L. Lin, *J. Nat. Gas Chem.* **2008**, *17*, 153–158.
- [29] J. Du, J. Yan, J. Hong, Y. Zhang, S. Chen, J. Li, *RSC Adv.* **2015**, *5*, 60534–60540.
- [30] A. Feller, M. Claeys, E. van Steen, *J. Catal.* **1999**, *185*, 120–130.
- [31] S. Ali, B. Chen, J. G. Goodwin, *J. Catal.* **1995**, *157*, 35–41.
- [32] B. Jongsomjit, J. Panpranot, J. G. Goodwin Jr., *J. Catal.* **2003**, *215*, 66–67.
- [33] J. Barrientos, V. Garcilaso, B. Veneza, A. Aho, J. A. Odriozola, M. Boutonnet, S. Jaras, *Top. Catal.* **2017**, *60*, 1285–1298.
- [34] W. Ma, V. R. R. Pendyala, P. Gao, T. Jermwongratanaichai, G. Jacobs, B. Davis, "Relating FTS Catalyst Properties to Performance, NASA/CR-2016-218485.
- [35] S.-J. Park, J. M. Cho, C.-I. Ahn, Y.-J. Lee, K.-W. Jun, B. G. Cho, J. W. Bae, *J. Mol. Catal. A* **2017**, *426*, 177–189.
- [36] J. W. Bae, S.-M. Kim, S.-H. Kang, K. V. R. Chary, Y.-J. Lee, H.-J. Kim, K.-W. Jun, *J. Mol. Catal. A* **2009**, *311*, 7–16.
- [37] S.-J. Park, J. W. Bae, Y.-J. Lee, K.-S. Ha, K.-W. Jun, P. Karandikar, *Catal. Commun.* **2011**, *12*, 539–543.
- [38] S.-J. Park, J. W. Bae, G.-I. Jung, K.-S. Ha, K.-W. Jun, Y.-J. Lee, H.-G. Park, *Appl. Catal. A* **2012**, *413–414*, 310–321.
- [39] M. H. Woo, J. M. Cho, K.-W. Jun, Y. J. Lee, J. W. Bae, *ChemCatChem* **2015**, *7*, 1460–1469.
- [40] J. W. Bae, S.-M. Kim, Y.-J. Lee, M.-J. Lee, K.-W. Jun, *Catal. Commun.* **2009**, *10*, 1358–1362.
- [41] M. K. Gnanamani, G. Jacobs, V. R. R. Pendyala, U. M. Graham, S. D. Hopps, G. A. Thomas, W. D. Shafer, D. E. Sparks, Q. Xiao, Y. Hu, B. H. Davis, *Appl. Catal. A* **2016**, *523*, 146–158.
- [42] M. C. J. Bradford, M. Te, A. Pollack, *Appl. Catal. A* **2005**, *283*, 39–46.
- [43] M. K. Gnanamani, G. Jacobs, U. M. Graham, V. R. R. Pendyala, M. Martinelli, A. MacLennan, Y. Hu, B. H. Davis, *Appl. Catal. A* **2017**, *538*, 190–198.
- [44] A. Y. Khodakov, A. Griboval-Constant, R. Bechara, V. L. Zholobenko, *J. Catal.* **2002**, *206*, 230–241.
- [45] D. D. Smiley, V. Gloriot, G. Jacobs, V. R. R. Pendyala, U. M. Graham, W. Ma, W. Shafer, G. A. Thomas, S. D. Hopps, A. MacLennan, Y. Hu, S. Khalid, B. Davis (2006) In: Fischer-Tropsch synthesis, catalysts and catalysts: advances and applications (B. H. Davis, M. L. Occelli, eds) vol 19, CRC Press Taylor & Francis Group, Boca Raton, FL, pp 55–83.
- [46] N. Okude, M. Nagoshi, H. Noro, Y. Baba, H. Yamamoto, T. A. Sasaki, *J. Electron Spectrosc.* **1999**, *101–103*, 607–610.
- [47] J. Prietzel, A. Dümig, Y. Wu, J. Zhou, W. Klysubun, *Geochim. Cosmochim. Acta* **2013**, *108*, 154–171.
- [48] J. S. Girardon, E. Quinet, A. Griboval-Constant, P. A. Chernavskii, L. Gengembre, A. Y. Khodakov, *J. Catal.* **2007**, *2*, 143–157.
- [49] R. L. Espinoza (Sastech), U. S. Patent 5733839, **1998**.
- [50] S. Vada, A. Hoff, E. Ådnanes, D. Schanke, A. Holmen, *Top. Catal.* **1995**, *2*, 155–162.

Manuscript received: April 11, 2018
Version of record online: August 1, 2018